

Nguyen Thai-Nghe
Thanh-Nghi Do
Salem Benferhat (Eds.)

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Proceedings, Part I

Part 1

Nguyen Thai-Nghe · Thanh-Nghi Do ·
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Editors

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Nguyen Thai-Nghe 
Campus II
Can Tho University
Can Tho, Vietnam

Thanh-Nghi Do 
Campus II
Can Tho University
Can Tho, Vietnam

Salem Benferhat 
Faculté des Sciences Jean Perrin
University of Artois
Lens, France

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Breast Cancer Classification Using Ensemble Deep Learning with RIDOPT-Net

Ho-Dat Tran^{1,2}, Thuong-Cang Phan¹, Van-Kha Vo³,
and Anh-Cang Phan²

¹ Can Tho University, Can Tho, Vietnam
ptcang@cit.ctu.edu.vn

² Vinh Long University of Technology Education, Vinh Long, Vietnam
{datth,cangpa}@vlute.edu.vn

³ Can Tho Oncology Hospital, Can Tho, Vietnam

Abstract. Early diagnosis and treatment of cancer are important steps in improving patient outcomes. In recent years, deep learning models have significantly assisted the medical field, improving the diagnostic and treatment capabilities of clinicians. However, individual models often produce inconsistent results and are sensitive to variations in input data, which can affect the classification performance of machine learning systems. In this paper, we propose the RIDOPT-Net model, which is designed to improve classification performance by integrating multiple machine learning models and optimizing their weights to reduce the classification error rate. The input data undergoes pre-processing to remove noise, normalize values, and enhance image quality before training. Based on each model's performance, the weights are automatically optimized using Optuna, enabling more effective ensemble integration. The proposed approach achieves a classification accuracy of 97.68% on the test dataset, demonstrating improved robustness and reliability. This study contributes to the development of a reliable and efficient breast cancer early detection system using intelligent systems and advanced data processing techniques in important healthcare applications.

Keywords: Breast cancer · mammogram · ensemble deep learning

1 Overview

In women, breast cancer is a disease with a very high incidence and mortality rate. For effective treatment, mammography plays an important role in timely screening for abnormalities. This technique is used for routine screening purposes for women diagnosed when there are suspicious clinical signs. Deep learning models have recently proven effective in medical image classification. However, combining multiple models further enhances accuracy and reduces classification errors. Eroğlu et al. [2] proposed a hybrid CNN system based on ultrasound images to diagnose BC by extracting features from Alexnet, MobilenetV2 and

Resnet50. The mRMR feature selection method was used to select the best features of the three models. This system uses support vector machine (SVM) and k-nearest (k-NN) machine learning algorithms as classifiers. The research results achieved an accuracy rate of 95.6%. Adedigba et al. [3] evaluated many deep learning models (InceptionV3, ResNet50, AlexNet, and CNN-TV-SVM ensemble method) on small-sized datasets, the proposed general method achieved an accuracy of 93.6%, although the efficiency of the proposed model is high, the small size of the dataset easily leads to a high possibility of overfitting when applied on many different datasets. Author Hashim [4] proposed a Hard Voting ensemble model to classify cancer on mammograms. This model combines three machine learning algorithms: Decision Tree, Logistic Regression and Support Vector Machine. Before training, the authors used Random Oversampling (ROS) to balance and normalize with Standard Scaler. The proposed model achieved a hard voting result of 98.25%, with an average accuracy over 10 cross-validation times of 97.38%. This study demonstrates that hard voting can provide high classification performance when the component models are selected and combined appropriately. Hanis et al. [5] conducted a meta-analysis to evaluate the accuracy of machine learning models in breast cancer classification from digital mammography and tomosynthesis. The study synthesized 36 studies with 68 machine learning models, using data from publicly available databases such as MIAS, DDSM, and primary data. The results showed high overall accuracy with an AUC of 0.90, sensitivity of 0.83, and specificity of 0.84. Factors affecting performance included country, data source, and model type, with neural network models and data from the United States performing best. However, the study also pointed out some limitations such as heterogeneity between studies and lack of tomosynthesis data. These findings support the potential application of machine learning in breast cancer screening.

This work trains multiple machine learning models to evaluate their classification performance, and then applies ensemble methods to leverage the strengths of each model, thereby enhancing the reliability of the classification results. Furthermore, we optimize the parameters based on Optuna techniques to enhance the performance during hard/soft voting and evaluate the proposed method against previous methods to validate the model performance.

2 Background

2.1 Breast Cancer

Mammography is a special X-ray method that uses a very small amount of X-rays to capture detailed images of the mammary gland. It helps detect characteristics of breast masses, especially microcalcifications, even when the mass cannot be felt through examination. Mammography has become an essential routine examination for protecting women's health in developed countries (Fig. 1).

Abnormalities on mammograms:

- **Mass:** Masses that increase density may represent benign or malignant masses. Benign masses often have clear boundaries and uniform density, while malignant masses usually have unclear boundaries and uneven density [1].

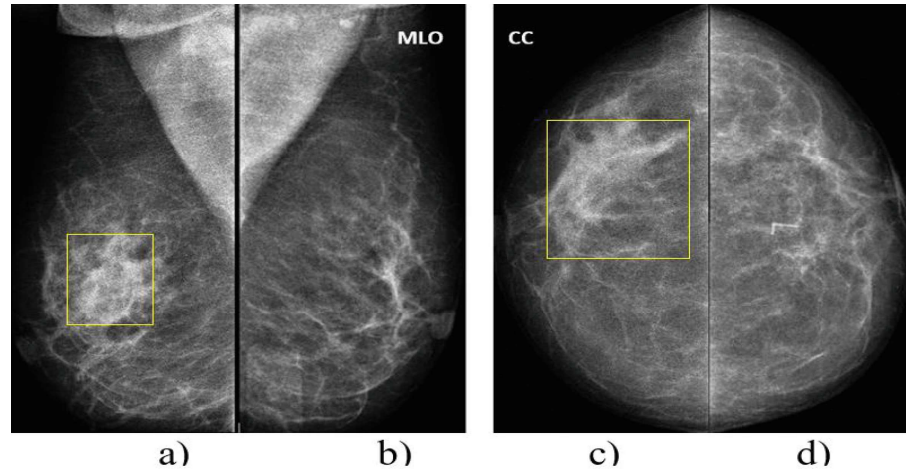


Fig. 1. Images on 2 views a) RMLO b) LMLO c) RCC d) LCC with the mass location localized in pictures a) and c)

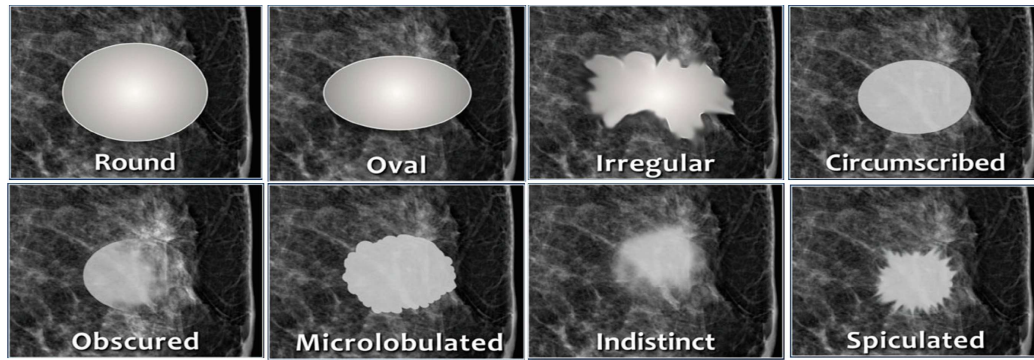


Fig. 2. Shape and margin of the lesion on mammogram [1]

- **Microcalcifications:** These are small calcium plaques that appear on mammograms as bright white spots or tiny dots on the soft tissue of the breast, as calcium easily absorbs X-rays during mammography. Microcalcifications are classified based on their shape, size, and location on mammograms [1] (Fig. 2).

2.2 Neural Network Models

InceptionV3 [6] is an enhanced iteration of the Inception architecture, derived from GoogLeNet, although substantially refined for feature extraction and computational efficiency. A significant enhancement is the utilization of factorized convolutions, substituting big filters (e.g., 5×5 or 7×7) with a sequence of

smaller filters ($1 \times n$ and $n \times 1$). This method maximizes parameter efficiency while maintaining the acquisition of intricate characteristics. Moreover, InceptionV3 employs Batch Normalization to accelerate convergence and mitigate overfitting, while using optimal pooling to diminish spatial dimensions. This architecture is appropriate for multi-class classification tasks due to its capacity to equilibrate depth and performance. ResNet152 [7] is a prominent variation of the Residual Network (ResNet) architecture that address the gradient vanishing issue in exceedingly deep networks by the use of skip connections. ResNet152, with 152 layers, employs bottleneck residual blocks (1×1 , 3×3 , 1×1 convolutions) to reduce dimensionality before applying the 3×3 convolution, thereby conserving memory. The ResNet152 architecture is appropriate for extensive datasets, such as ImageNet, due to its capacity to learn profound hierarchical features. ResNet50V2 enhances the original ResNet by reorganizing the configuration of Batch Normalization and skip connections. ResNet50V2 positions batch normalization prior to the pre-activation function rather than following each convolutional layer, facilitating more stable gradient propagation, enhancing accuracy, and optimizing the balance between depth (50 layers) and training speed, making it appropriate for real-time applications. DenseNet121 [8] is built on the principle of Dense Connectivity, wherein each layer obtains input from all preceding levels via concatenation. This facilitates the efficient reuse of features throughout the network, minimizes parameters through inter-layer information sharing, and mitigates vanishing gradients by providing several backpropagation pathways for gradients. DenseNet121, comprising 121 layers, is appropriate for resource-limited scenarios necessitating high accuracy, whereas DenseNet201, having 201 layers, improves the capacity to learn intricate features, particularly in medical imaging such as skin lesion or X-ray categorization.

2.3 Ensemble Deep Learning

Ensemble Deep Learning [10] based on the combination of many different deep learning models helps to improve the accuracy and stability of classification results. By taking advantage of the unique advantages of each architecture such as the diverse feature extraction of InceptionV3, the stability of ResNet, or the parameter efficiency of DenseNet. Using model ensemble helps to reduce variance, limit overfitting and improve generalization ability. Two popular techniques for combining model outputs are hard voting and soft voting. Given a set of machine learning models $m_1, m_2, \dots, m_i$. Each model m_i predicts for sample x in the form of label $m_i(x)$ belonging to $0, 1$ or probability $p_i(c|x)$, with c belonging to $0, 1$. Each model has a corresponding weight $w_i \geq 0$ representing the prediction confidence on the validation set. The goal of the two soft voting and hard voting functions is to determine the final predicted label t for sample x through the combination of machine learning models.

Hard Voting: Hard voting [11] is a simple and popular model ensemble technique, often used in classification problems. This method is based on the majority voting mechanism: each component model in the ensemble will independently predict the label for an image as benign or malignant based on the majority vote from the models.

$$\hat{t}_{hard} = \text{mode}(\{m_1(x), m_2(x), \dots, m_i(x)\}) \quad (1)$$

$$\hat{t}_{weight.hard} = \arg \max_c \left(\sum_{i=1}^k w_i \cdot 1_{\{m_i(x)=c\}} \right) \quad (2)$$

In which, w_i : weight of the i^{th} model; $1_{\{m_i(x)=c\}}$: equal to 1 if model i chooses class c , otherwise equal to 0 (Fig. 3).

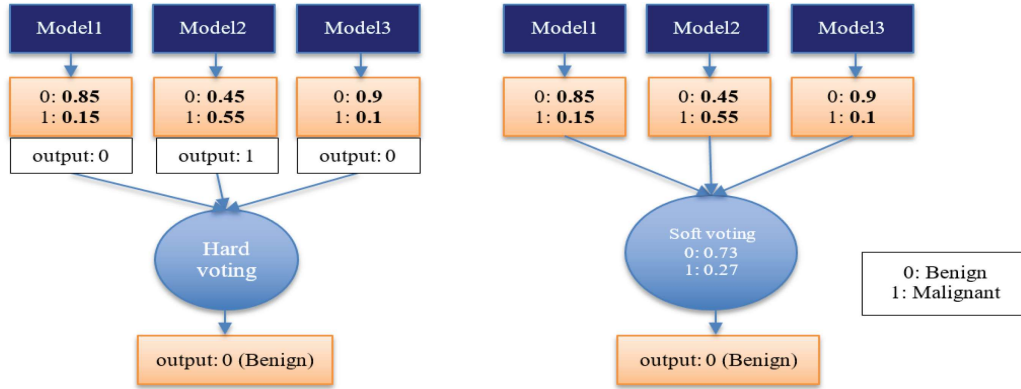


Fig. 3. Illustration of the working mechanism of hard voting and soft voting.

Soft Voting. Soft voting [11] is an advanced model ensemble method, often applied when the component models are capable of producing predicted probabilities or scores instead of just giving classification labels. The working principle of soft voting is to average the predicted probabilities from the component models to decide the final label. The general formula of the soft voting and weighted soft voting method is presented as follows:

$$\hat{t}_{soft} = \arg \max_c \left(\frac{1}{n} \sum_{i=1}^n p_i(c|x) \right) \quad (3)$$

$$\hat{t}_{weight.soft} = \arg \max_c \left(\sum_{i=1}^n w_i \cdot p_i(c|x) \right) \quad (4)$$

In which: n : number of classification models. $p_i(c|x)$: probability that the model predicts sample x to belong to class c .

2.4 Evaluation Metrics

We use some common metrics in machine learning to evaluate the performance of the model for detecting and classifying breast cancer on mammograms. These metrics include accuracy based on total number of experimental images, sensitivity, F1-score, and model loss during training. The formulas used in the paper include:

$$Accuracy = \frac{True.Pos + True.Nega}{True.Pos + True.Nega + Fal.Pos + Fal.Nega} \quad (5)$$

$$Precision = \frac{True.Pos}{True.Pos + Fal.Pos} \quad (6)$$

$$Recall = \frac{True.Pos}{True.Pos + Fal.Nega} \quad (7)$$

$$F1 - score = 2 \times \frac{Pre. \times Rec.}{Pre. + Rec.} \quad (8)$$

$$Loss(C, P) = - \sum_i C_i \log(P_i) \quad (9)$$

$$mAP = \frac{1}{n} \sum_{i=1}^{i=n} AP_i \quad (10)$$

Where, True.Pos: Number of benign samples correctly identified by the model; True.Nega: Number of malignant samples correctly identified by the model; Fal.Pos: Number of benign samples mistakenly identified as malignant by the model; Fal.Nega: Number of malignant samples mistakenly identified as benign by the model. C_i : Correct label of the i^{th} class; P_i : Probability of class i predicted by the model; n : the number of classes; AP_i : Average precision for class i .

3 Proposed Method

In this paper, we introduce a three-stage design for breast cancer classification. The first stage involves preprocessing, where raw mammogram images are enhanced and augmented to generate suitable inputs for deep learning. The second stage focuses on feature extraction and model training using multiple CNN architectures. Finally, Optuna-based weight optimization is applied to integrate the models through hard and soft voting, followed by the final classification into benign or malignant categories. The overall workflow of the proposed model is illustrated in Fig. 4.

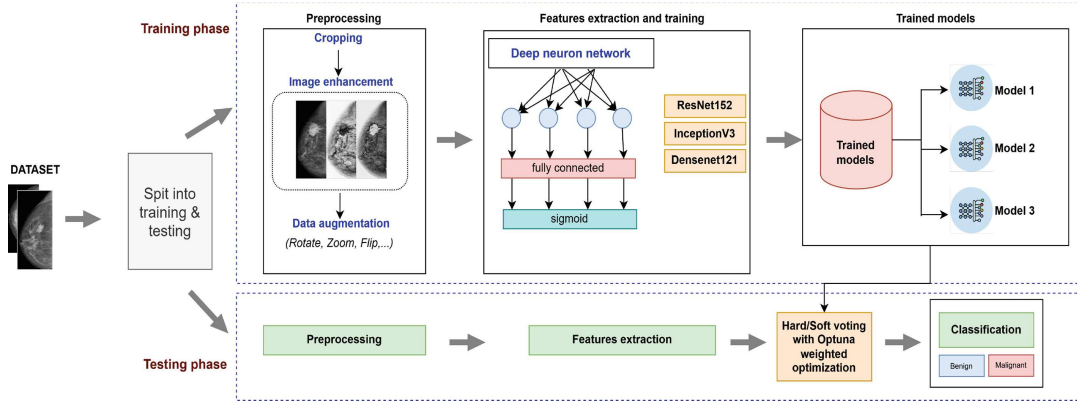


Fig. 4. RIDOPT-Net model in cancer classification on mammography.

3.1 Training Phase

In the training phase, the input data is preprocessed using a number of techniques such as cropping, image enhancement, and data augmentation (flip, rotate, zoom, etc.). Next, the images are fed into the proposed machine learning models for feature extraction and training. The dataset is trained with 3 separate deep learning models including: InceptionV3, ResNet152, DenseNet121. Each model is trained independently on the same dataset and unified parameters. We use the early stopping technique during training to limit the model from overfitting. The best training results of each model are saved for testing.

3.2 Testing Phase

The test image is processed similarly through data preprocessing steps, feature extraction, then hard voting and soft voting techniques are applied, combined with weights optimized by Optuna technique on 3 models, to give the final result as benign or malignant.

3.3 Optuna Weighted Optimization

The RIDOPT-Net algorithm is an ensemble learning method that uses Optuna weight optimization to improve image classification performance.

Algorithm 1. RIDOPT-Net algorithm

```

1: Input: Trained models  $\{m_1, m_2, \dots, m_i\}$ , test data  $X_{\text{test}}$ , true labels  $y_{\text{test}}$ , voting
   type  $\text{mode} \in \{\text{soft}, \text{hard}\}$ 
2: Load model outputs for each  $m_i$ .
3: Initialize Optuna study: maximize accuracy
4: for each trial in Optuna do
5:   Sample weights  $w_1, w_2, \dots, w_m \in [0, 1]$ 
6:   Normalize:  $w_i \leftarrow \frac{w_i}{\sum_{j=1}^m w_j}$ 
7:   if  $\text{mode} = \text{hard}$  then
8:     Compute ensemble probability using Eq.2
9:     Predict label
10:  else if  $\text{mode} = \text{soft}$  then
11:    Compute weighted vote using Eq.4
12:    Predict label
13:  end if
14:  Compute accuracy:  $\text{Acc} = \text{accuracy}(y_{\text{test}}, \hat{t})$ 
15:  Report  $\text{Acc}$  to Optuna
16: end for
17: Extract best weights  $w_i^*$  from best trial
18: return Final prediction  $\hat{t}$  using  $w_i^*$ 
19: Output: Optimal weights  $w_i^*$  and final prediction  $\hat{t}$ 

```

The algorithm takes as input a set of trained deep learning models, test data, real-world labels, and a voting mechanism (soft or hard). It then predicts on the test set using each model and initiates an optimization session with the goal of optimizing accuracy. In each trial, the algorithm generates a random combination of weights for the models, normalizes them, and then applies them to calculate the voting result. If it is soft voting, the probabilities from the models are weighted to make a prediction; if it is hard voting, the binary predictions from each model are weighted. The prediction accuracy is computed and reported to Optuna, which identifies the optimal weight combination. Finally, the algorithm outputs the optimized weights along with the final prediction. RIDOPT-Net has the advantage of automating the process of finding the best weights, flexibly combining multiple models, and supporting both soft and hard voting, which significantly improves accuracy in classification systems. We perform experimental scenarios based on the suggested algorithm to assess the method’s efficacy and offer an analysis of the experimental findings (Table 1).

4 Experimental Results

4.1 Installation Environment and Experimental Dataset

The system is installed in Python and runs on the Kaggle environment with an Intel®Xeon®2.00 GHz processor, 30GB RAM, and a graphics processor with 16GB RAM configuration, using an Nvidia Tesla P100 GPU. The library used to support network model training is TensorFlow v1.5.

Table 1. Recommended scenarios and training parameters

Scenario (Sce.)	Name	Dimension (H×W)	Batch Size	Epochs	Optimizer	Learning Rate
1	InceptionV3	384×192	32	100	Adam	1e-4
2	Resnet152	384×192	32	100	Adam	1e-4
3	Densenet121	384×192	32	100	Adam	1e-4
4	Ensemble Deep Learning (Hard voting)					
5	Ensemble Deep Learning (Soft voting)					
6	RIDOPT-Net (Hard voting)					
7	RIDOPT-Net (Soft voting)					

Dataset: The dataset was collected from the VinDr dataset [11] and several hospitals, and then reviewed by physicians to determine benign and malignant classification. After preprocessing the slices containing the lesions and converting them to .jpg format, the training and testing datasets consisted of 6940 images. Of these, 5862 images were used for training and 1078 images for testing, corresponding to a ratio of 80/20.

4.2 Training Results

The training and testing outcomes of deep learning models applied to the medical image classification problem using the suggested model are shown in this section. We use evaluation metrics like Precision, Recall, F1-score, and Accuracy for each distinct class to assess the model on the test set throughout the training phase. To assess each model's efficacy, separate models like DenseNet121, InceptionV3, and ResNet152 are trained and tested separately. We compare the effectiveness of each model based on the results on the level of recognition of two classes. We then apply the Optuna weight optimization method to combine the models together. Each model's weights are automatically adjusted to enhance the system's overall performance. To assess the ensemble method's enhanced performance in the medical image classification challenge, the optimum weighting scenarios are compared holistically (Fig. 5).

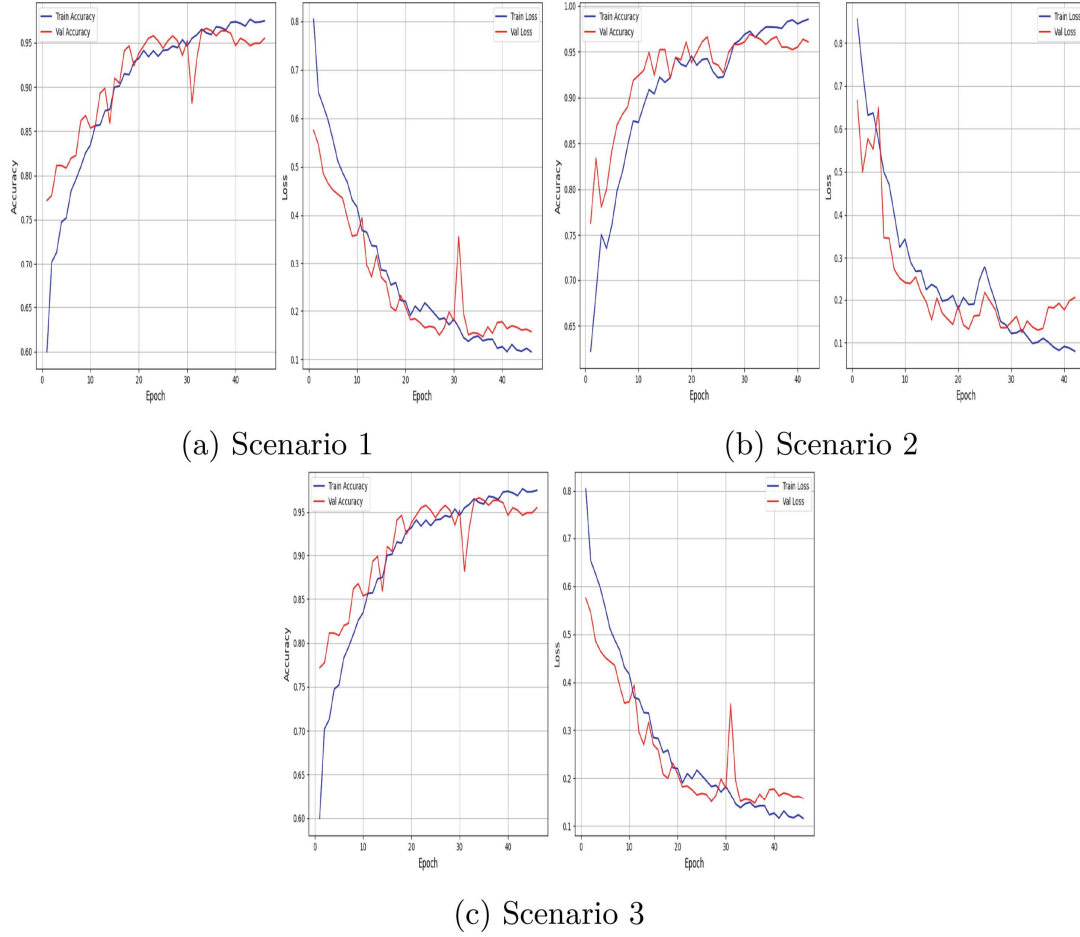


Fig. 5. Accuracy and Loss measurement in three neural network models.

Figure 6 shows that all three models achieve high training accuracy: Sce. 1 reaches the highest (96.5%), followed by Sce. 3 (95.6%), while Sce. 2 performs the lowest (94.1%). In terms of training error, Sce. 3 gives the best result with the lowest loss of 0.184, while Sce. 2 has the highest loss of 0.209. This shows that although Sce. 1 has higher accuracy, Sce. 3 is more optimal in terms of loss. In terms of training time, Sce. 2 is the fastest with 202.14 min, while Sce. 3 takes the longest time of 244.89 min. Overall, Sce. 3 shows a good balance between high accuracy, low loss and training time, while Sce. 1 stands out in terms of accuracy but is not as optimal in terms of time and loss as Sce. 3. Based on the advantages of each model, we conduct integration Sce. 4, 5, 6, 7 to test the effectiveness.

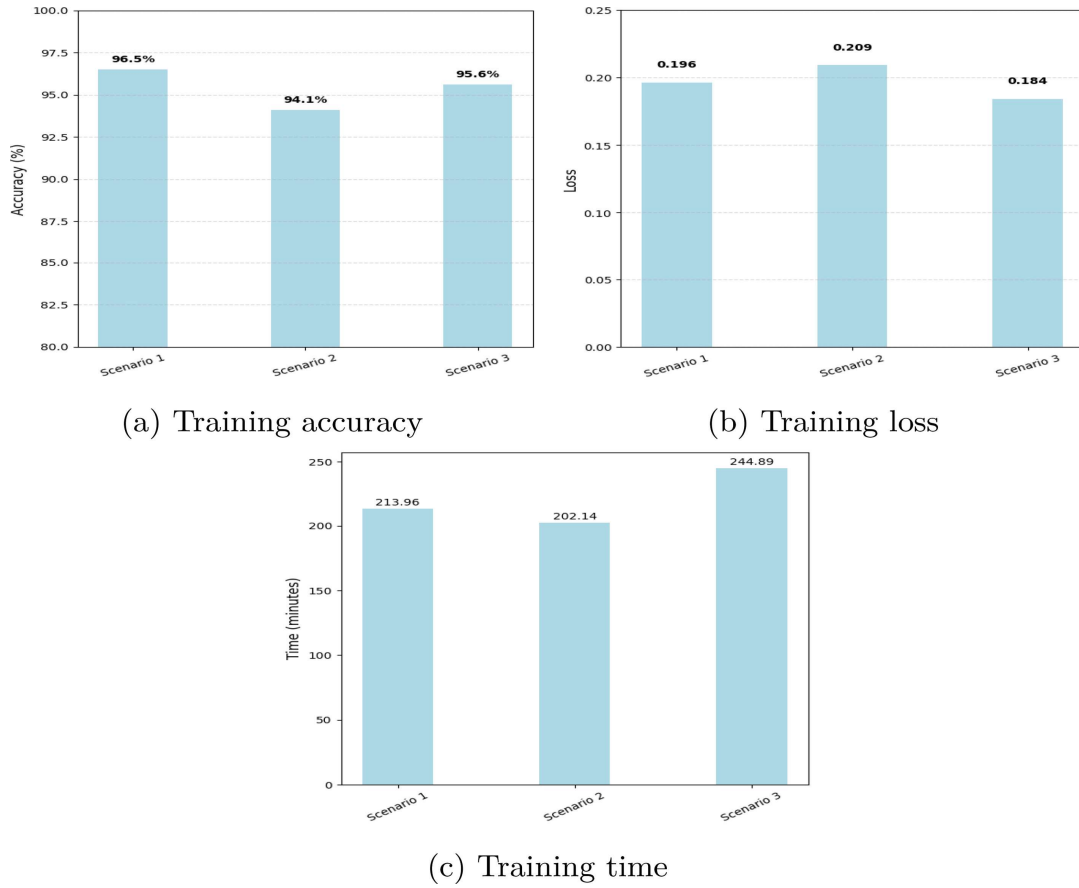


Fig. 6. Training results of 3 proposed models.

In addition to evaluating the results of each model, we are interested in the performance contribution of each model when conducting the combination. The evaluation results include the correlation between the models and the level of importance of the model's contribution, the results are presented in Fig. 8 and Fig. 9 (Fig 7).

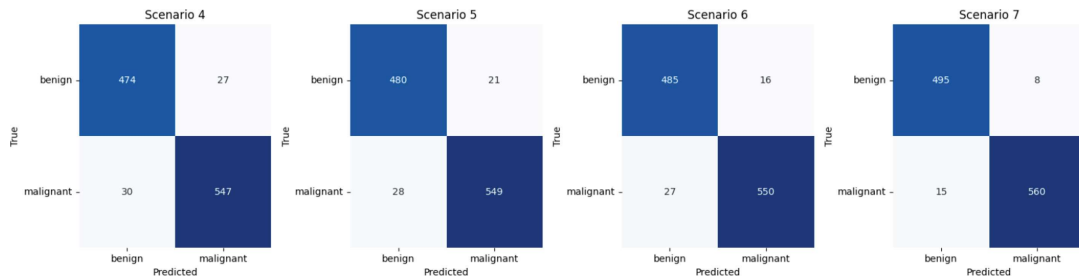


Fig. 7. Confusion matrix of scenarios 4, 5, 6 and 7.

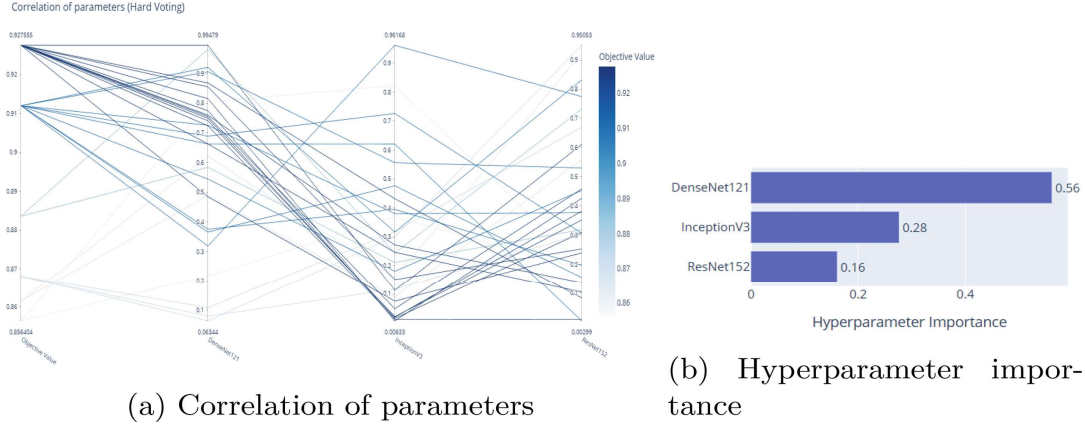


Fig. 8. Correlation and importance of parameters in Sce. 6.

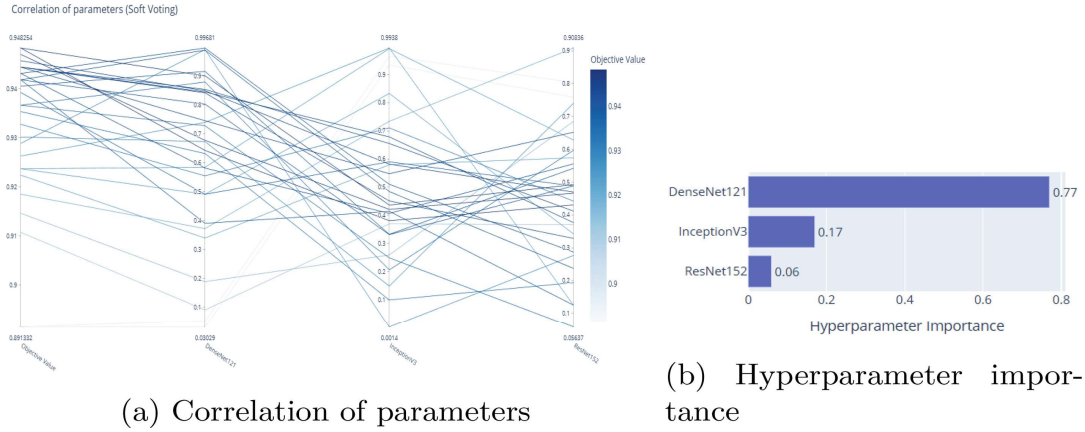


Fig. 9. Correlation and importance of parameters in Sce. 7.

Figure 8 and Fig. 9 shows the correlation and importance of the models in the parameter optimization process in Sce. 6, 7. In figure (a), the graph shows the correlation between the Objective Value and the weights of three models: DenseNet121, InceptionV3, and ResNet152. The colors of the lines represent the quality of each weight combination—darker lines correspond to higher objective values. The results show that the best objective values are often associated with high weights for DenseNet121, while InceptionV3 and ResNet152 often have lower weights. In both scenarios, DenseNet121 has a large influence on classification when combined with 3 models, this is the model that plays a key role in improving the accuracy on the test set. Meanwhile, InceptionV3 and ResNet152 only have a low proportion, indicating that their contribution is quite limited (Table 2).

Table 2. Evaluation results on test dataset.

Scenario	Classify	Precision(%)	Recall(%)	F1-score(%)	mAP(%)
Sce. 1	B	94.62	94.67	92.04	94.61
	M	94.21	94.93	91.17	
Sce. 2	B	96.53	94.05	95.32	95.63
	M	94.81	97.12	95.91	
Sce. 3	B	95.33	92.03	93.63	94.12
	M	93.21	95.91	94.52	
Sce. 4	B	94.61	94.05	94.33	94.69
	M	94.80	95.30	95.05	
Sce. 5	B	95.81	94.49	95.14	95.44
	M	95.15	96.32	95.73	
Sce. 6	B	96.81	94.73	95.76	96.01
	M	95.32	97.17	96.24	
Sce. 7	B	98.41	96.68	97.54	97.68
	M	97.05	98.59	97.82	

The results table shows the classification performance between the two classes Benign (B) and Malignant (M) in seven different scenarios, with the measures including Precision, Recall, F1-score and Accuracy. The results show that Sce. 7 achieves the highest performance with Accuracy up to 97.68%, with Precision, Recall and F1-score being superior in both classes (B: 98.41% Precision, M: 97.82% F1-score). This shows that using the Optuna technique in weight optimization is the optimal solution in accurately classifying between the two classes. Meanwhile, Sce. 1 to 6 have accuracy ranging from 94.12% to 96.01%. In which, Sce. 6 stands out with Accuracy 96.01%, showing quite high efficiency, especially with F1-score class B reaching 95.76% and class M reaching 96.24%. Scenarios such as 3 and 4 have lower Accuracy (94.12% and 94.69% respectively), showing that the classification ability is not optimal. Through the synthesis results, it can be seen that the proposed model RIDOPT-Net by combining the advantages of each model and combining the weight optimization method brings outstanding performance in classification on mammograms (Fig. 10).

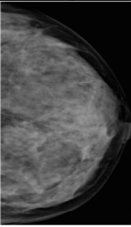
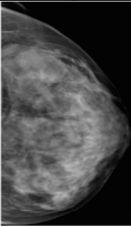
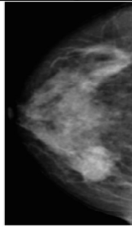
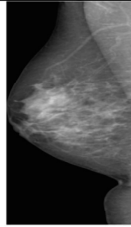
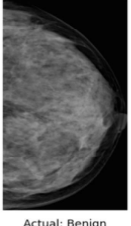
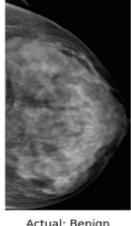
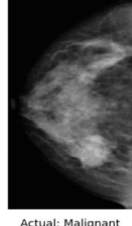
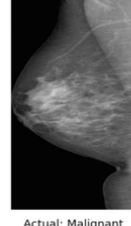
Method	Output			
	Img1	Img2	Img3	Img4
Scenario 6				
	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign
Scenario 7				
	Actual: Benign Predict: Benign p soft(c=0): 1.00 p soft(c=1): 0.00	Actual: Benign Predict: Benign p soft(c=0): 1.00 p soft(c=1): 0.00	Actual: Malignant Predict: Malignant p soft(c=0): 0.00 p soft(c=1): 1.00	Actual: Malignant Predict: Malignant p soft(c=0): 0.00 p soft(c=1): 1.00

Fig. 10. Prediction results on the test dataset of Sce. 6 and 7.

5 Conclusion

In this paper, we propose a comprehensive method for detecting and classifying breast cancer using deep learning ensemble models. The input image is preprocessed using a series of image enhancement techniques such as rotation, translation, zooming, and flipping. These techniques help create many different copies of the original image, thereby increasing the size and diversity of the training dataset, which is important for effective model training. We propose RIDOPT-Net, a deep learning ensemble framework that leverages the complementary strengths of multiple models to improve classification performance. In addition, we perform optimization among the Optuna techniques to help the model automatically optimize the voting process in both hard and soft methods. The proposed model improves classification on both classes compared to the baseline model and helps to increase the classification efficiency on both classes, achieving 97.68% on the experimental dataset. In the future, we continue to perform comparisons on various network models and perform on several segmentation models, apply testing on multiple datasets, and address big data challenges to better accommodate the growing volume of data in medical imaging.

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